National 5 Physics

Space, Waves, & Radiation

Rapid-Fire Paper Marking Scheme

Section 1 — Space

Question			Expected response	Max mark	Additional guidance
4.	(a)		Hydrogen, helium and mercury	1	Must have all three.
	(b)	(i)	The distance light travels in one year.	1	
		(ii)	$d = vt$ (1) $d = 3{\cdot}0 \times 10^8 \times (60 \times 60 \times 24 \times 365{\cdot}25 \times 97)$ (1) $d = 9{\cdot}2 \times 10^{17}$ (m) (1)	3	Calculation can be carried out in steps, but all steps must be done for the substitution mark to be awarded, eg calculation of distance for one light-year, followed by multiplying this by 97. Unit in final answer not required, but if stated, must be correct. Accept 1-4 sig fig: 9×10^{17} $9{\cdot}18 \times 10^{17}$ $9{\cdot}183 \times 10^{17}$ Also accept, if using 365 days: $9{\cdot}177 \times 10^{17}$
	(c)	(i)	No atmosphere to absorb light full range of EM waves can be observed can be used in cloudy weather/ daytime no light pollution	1	Or other suitable response Do not accept 'closer' but would not negate a correct response. Apply +/- rule for surplus answers
		(ii)	GPS weather forecasting communications scientific discovery	1	Or other suitable response Do not accept responses relating to space observation Apply +/- rule for surplus answers

	(b)	(i)	Solar cells	1	Accept: solar panels Radioisotope Thermoelectric Generator (RTG) nuclear reactors or other suitable answer Solar energy/power alone is insufficient. Nuclear energy/power/reactions alone is insufficient. (Rechargeable) batteries/cells alone is insufficient.
		(ii)	Manoeuvring in zero friction environment **OR** Fuel load on take-off **OR** Potential exposure to radiation **OR** Pressure differential **OR** Re-entry through an atmosphere	1	Accept any other suitable answer. Do not accept: 'it takes a long time' alone 'cost'

12.	(a)	(i)	$3{\cdot}0 \times 10^{8}$ m s^{-1}	1	Accept: 3×10^{8} m s^{-1} 300 000 000 m s^{-1}
		(ii)	$d = vt$ (1) $d = 3{\cdot}0 \times 10^{8}$ $\times(7{\cdot}8 \times 365{\cdot}25 \times 24 \times 60 \times 60)$ (1) $d = 7{\cdot}4 \times 10^{16}$ (m) (1)	3	Accept 1-4 sig fig: 7×10^{16} (m) $7{\cdot}38 \times 10^{16}$ (m) $7{\cdot}384 \times 10^{16}$ (m) Also accept, if using 365 days: $7{\cdot}379 \times 10^{16}$ (m)
	(b)	(i)	photographic film	1	Accept: 'charge coupled device'/'CCD' 'photodiode' 'phototransistor' 'retina (of the eye)' 'LDR'
		(ii)	equal (to)	1	Accept equivalent statement (eg 'same')

Section 2 – Waves

Question			Answer	Max Mark	Additional Guidance
4.	(a)		(black bulb) thermometer, photodiode, phototransistor, thermistor, thermocouple, CCD, thermochromic film	1	Do not accept: Skin (Infrared/thermal imaging) camera Photographic film thermogram
	(b)		Gamma (radiation/rays)	1	
	(c)	(i)	** SHOW THAT ** Must start with the correct equation or (0) $v = f\lambda$ (1) $3{\cdot}0 \times 10^8 = 1{\cdot}2 \times 10^9 \times \lambda$ (1) $\lambda = 0{\cdot}25$ m	2	Final answer of 0·25 m or its numerical equivalent, including unit, must be shown, otherwise a maximum of (1) can be awarded. For alternative methods calculating v or f there must be final statement to show that calculated value of v or f is the same as the value stated in the question/data sheet to gain the second mark.
		(ii)	Microwave (radiation)	1	Accept: 'microwaves'

Question			Answer	Max Mark	Additional Guidance
6.	(a)	(i)	normal drawn <u>and labelled</u>	1	Must be 'passably' perpendicular and straight Does not need to be dashed Accept: 'N' or 'n' as label
		(ii)	Both angles indicated and labelled	1	Accept: i and r I and R θ_i and θ_r If normal has been incorrectly drawn, then this mark is still accessible, provided angles are indicated to the normal and labelled. Accept angles indicated either entering or leaving the Perspex block
	(b)	(i)	8°	1	Allow ±0·5° tolerance Unit must be included
		(ii)	Any single value between 40° and 42° inclusive.	1	Unit must be included
	(c)		Any one of: To obtain more reliable results Eliminate rogue results/outliers To allow an average/mean to be calculated More accurate	1	Do not accept: 'more precise' 'better results' 'to make it a fair test'

Section 3 – Radiation

			Answer	Max Mark	Additional Guidance
6.	(a)		The time taken for the activity / corrected count rate (of a radioactive source) to half.	1	Do not accept: Time for radiation / radioactivity / count rate to half.
	(b)	(i)	Measure the count in a set time interval (1) Repeat at (regular) intervals (1) Measure background (count) and subtract (1)	3	(3) independent marks. Description must refer to the apparatus shown. If candidate response makes reference to using a rate meter then MAX (2) marks.
	(b)	(ii)	(Half-life =) 10 minutes (1)	1	Unit required (accept mins) +/- half box tolerance
		(iii)	88 ⟶ 44 ⟶ 22 ⟶ 11 ⟶ 5·5 (1) mark for evidence of halving Count rate = 5·5 counts per minute (1)	2	Or answer consistent with 6(b)(ii) Accept 5 or 6 counts per minute Accept calculation based on one halving of 11 counts per minute Unit required (accept c.p.m.) Alternative method: Accept calculation using division by 2^4 (equivalent to halving).

Question			Answer	Max Mark	Additional Guidance
8.	(a)	(i)	$D = \frac{E}{m}$ (1) $= \frac{7 \cdot 2 \times 10^{-3}}{80 \cdot 0}$ (1) $= 9 \cdot 0 \times 10^{-5}$ Gy (1)	3	
		(ii)	$H = Dw_R$ (1) $= 9 \cdot 0 \times 10^{-5} \times 1$ (1) $= 9 \cdot 0 \times 10^{-5}$ Sv (1)	3	Or answer consistent with 8(a)(i) If wrong radiation weighting factor selected then (1) MAX for correct equation.
	(b)		When an atom gains / loses / gains or loses electrons.	1	Ignore additional information.

[END OF MARKING INSTRUCTIONS]